



Selenium Course Content (12 days)

Core Java, HTML & CSS – 2 days

Core Java :

Introduction to all predefined packages

More about JVM, JRE & JDK

Types of variables

Data Types in Java

Introduction to OOPS

- Class

- Object

- Local, Instance and static variables

- Constructors

- Methods

- Arrays

- This & super keyword

- Final Keyword

- Inheritance and Types

- Working with super classes

- Using types of polymorphism such as overloading, overriding, and dynamic binding

- Abstraction

- Encapsulation

- Abstract Classes

- Interfaces

String Methods

Collections – ArrayList, Set, Map

Exception Handling

IO Streams Basics

HTML & CSS

Selenium : - 6 days

- Introduction to Automation

- Introduction to Selenium

 - Selenium Overview

 - Features of Selenium

- Maven Overview

 - Introduction To Maven Goals

 - Creating a Maven Project

 - Setting up Maven project with Selenium Dependency

- Selenium WebDriver 4 API

 - Introduction to Selenium WebDriver

 - WebDriver Architecture



Interacting with WebElements

Button, TextBox, TextArea, Dropdown, RadioButton, Checkbox, ToolTip

Locators

Locator Strategies - Id, name, linkText, PartialLink, tagName, cssselector, xpath, classname

Relative Locators

Dynamic xpath

Handling Alerts

Handling Multiple Windows

Handling Web Tables

Handling iFrames

Interacting with ShadowDOM

Synchronization

Using Javascript Executor

TestNG Framework

Introduction to TestNG framework

TestNG Annotations

Creating and configuring Testng.xml file

TestNG reports

Running failed testcases

Running TestNG test as Maven test

TestNG Options

Priority

Groups and Parent Group

Parallel Execution

Parameterization

Implementing DataDriven Framework

Using .CSV file as Dataprovider

Reading from External Files

Using property file

Using excel as Object Repository

Using .json as Data File

Extent Reports

Generating Extent Reports

Attaching screenshots with ExtentReports

Selenium Grid

Components in Selenium Grid

Running in Standalone and Hub&Node mode

Implementing Design Pattern – Page Object Model

Page Object Model

Page Factory

Implementation of POM & PF

Framework Design

Automating any eCommerce portal using POM by implementing a framework.



API Testing using RestAssured – 1 day
Rest API Testing - Get, POST, PUT, DELETE
Post using .json payloads
API request using Authentication

Version Control – 1 day

Working with Git and GitHub
Basic Git commands - init, add, commit, push, pull, clone

AI Driven

Configure Selenium MCP Server in Claude AI and generate testscripts.
Casestudy & Evaluation – 2 days



Playwright using TypeScript (12 days)

Prerequisites :

Web application manual testing knowledge is a must

Required Software :

NodeJS V22+
Typescript 5.2.2, Java 17
Npm
VS Code V1.101
GitBash
Chrome, Edge
Claude AI

Detailed Course Contents :

TypeScript Introduction & Environment Setup

Javascript - Callback functions, Promises, Async / Await
What is TypeScript?
Setting up Node.js & VS Code

Typescript, HTML & CSS – 3 days

Transpiling .ts to .js
Data types - string, number, boolean, any, void, union

Functions

Function declarations & expressions
Optional and default parameters
Rest parameters and return types
Arrow functions
Callback functions

Arrays & Tuples

Class Features

Creating classes
Readonly properties

Inheritance & Access Control

extends, method overriding
Access modifiers: public, private, protected
Importing/exporting modules

HTML & CSS

Playwright – 6 days

Introduction to Playwright
Installing Playwright
Initializing Playwright project
Understanding project structure
Writing first test
Running tests (headless/headed mode)
Running specific tests & multiple tests
UI Mode overview

Playwright Locators

getByText, getByRole, getByLabel



locator(), page.locator()

CSS Locators

Tag, ID, Class, Attribute selectors

Basic Actions

Typing text, clicking buttons

Handling radio buttons, checkboxes

Dropdown

Tooltips

iFrames

Locating elements inside iframes

Switching contexts

Capture Text of the Elements

innerText() vs.textContent()

allInnerTexts() vs.allTextContents()

all() Method

Accessing Web Tables

Dialogs, Windows & Tabs

Handling alerts, prompts, confirm dialogs

Switching between tabs/windows

Mouse Actions

Hover, right-click, double-click, drag & drop

Scrolling

Scrolling to elements

Infinite scrolling techniques

File Upload & download

Shadow DOM

Handling Shadow DOM & Nested Shadow DOM Elements

Browser Context & Pages

Working with new contexts

Multi-tab and isolated sessions

Auto-waiting, Timeouts & Assertions

Auto-waiting mechanism

Test and Expect Timeouts

Assertions (Auto-retrying & Non-retrying assertions)

Record the tests (Test Generator/Codegen)

Record the tests using Codegen

Identify locators in Codegen

Modifying generated tests

Debugging tests

Live debugging, editing locators

Run in Debug Mode

Picking a Locator

Screenshots & Videos

Capturing screenshots

Recording test runs

Viewport

Tracing with Trace Viewer



- Enabling tracing
- Opening trace files for debugging
- Retries and handle Flaky tests

Grouping Tests & Hooks

- grouping tests with describe()
- Hooks - beforeAll, afterAll, beforeEach, afterEach

Tagging & Annotations

- Annotations
- Tagging Tests

Parallel Test Execution

- Setting parallelism in config
- Worker-level isolation & Limiting workers

Parameterization

- Different ways to Parameterize data to the tests
- Data Driven Testing with External files(JSON)

Reporting

- Default reports

Page Object Model (POM)

- Creating reusable page classes
- Encapsulating page actions

Visual Testing/Visual Comparisons

- Generating Screen shots
- Comparing snapshots

API Testing with Playwright - 1 Day

- Rest API Testing - Get, POST, PUT, DELETE
- Post using .json payloads
- API request using Authentication

Version Control – 1 Day

- Working with Git and GitHub
- Basic Git commands - init, add, commit, push, pull, clone

AI Driven

- Configure Playwright MCP Server in Claude AI and generate testscripts.

Casestudy & Evaluation – 2 days